

JAVA PROGRAMMING SIDDHARTHA ACADEMY OF HIGHER EDUCATION (DEEMED TO BE  
LAB (23AI5351) UNIVERSITY) • DEPARTMENT OF AI • BATCH 2025–2029

## Week 08: Collections Framework & File Streams I/O

Student: Trivikram Roll No: 25EU02067 • Lab Record Verified

### 1. OBJECTIVE / AIM

To implement the Java Collections Framework (ArrayList, LinkedHashMap) and demonstrate character and byte file streams.

### 2. CONCEPT & THEORY

The Java Collections Framework provides standardized data structure abstractions. The java.io package enables stream-oriented reading and writing of binary and textual data.

### 3. JAVA SOURCE IMPLEMENTATION

```
// Java Programming Lab - Experiment 08
// Student: Trivikram | Roll No: 25EU02067
import java.util.*;
import java.io.*;

public class CollectionsAndIODemo {
    public static void main(String[] args) {
        // Collections Framework Demonstration
        Map<String, String> registry = new LinkedHashMap<>();
        registry.put("25EU02067", "Trivikram");
        registry.put("25EU02068", "Venkatesh");

        System.out.println("=== Student Lab Registry ===");
        registry.forEach((roll, name) -> System.out.println("Roll: " + roll + " | Stu

// Stream I/O Operations
String message = "Java Programming Lab 23AI5351 - Experiment 8 Completed.";
try (ByteArrayInputStream in = new ByteArrayInputStream(message.getBytes());
    ByteArrayOutputStream out = new ByteArrayOutputStream()) {

    int c;
    while ((c = in.read()) != -1) {
        out.write(Character.toUpperCase(c));
    }
    System.out.println("\nProcessed Stream Output:");
    System.out.println(out.toString());
}
```

```
        } catch (IOException e) {  
            e.printStackTrace();  
        }  
    }  
}
```

#### 4. COMPILATION & EXECUTION OUTPUT

```
=== Student Lab Registry ===  
Roll: 25EU02067 | Student: Trivikram  
Roll: 25EU02068 | Student: Venkatesh  
  
Processed Stream Output:  
JAVA PROGRAMMING LAB 23AI5351 - EXPERIMENT 8 COMPLETED.
```

